D.A.V. PUBLIC SCHOOL, SECTOR IV, B. S. CITY

CLASS XII — MATHEMATICS REVISION SHEET (SET 2)

QUESTION BANK VARIANT B **TOPIC-WISE STRUCTURED COMPILATION**

GENERAL INSTRUCTIONS:

1. This sheet provides a topic-wise compilation of Class XII Mathematics board-level question variants for final preparation and systematic review.
2. Solve problems in each section to solidify your understanding of Vectors, 3D Geometry, Linear Programming, and Bayes' Theorem applications.

UNIT 1: VECTOR ALGEBRA

Q1. If a is a non-zero vector of magnitude k and λ is a non-zero scalar, then (λa) is a unit vector when:

(a) $\lambda = k$ (b) $k = 1 / |\lambda|$

(c) $\lambda = |k|$ (d) $k = |\lambda|$

Q2. If vectors $\mathbf{u} = 4\mathbf{i} - p\mathbf{j} + 3\mathbf{k}$ and $\mathbf{v} = 3\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$ are mutually perpendicular, then p equals:

(a) 12 (b) 6

(c) 24 (d) 18

Q3. Given two vectors a and b such that $|\mathbf{a}| = 4$, $|\mathbf{b}| = 3$ and $\mathbf{a} \cdot \mathbf{b} = 6\sqrt{3}$, determine the angle θ between them:

(a) π/6 (b) π/3

(c) π/4 (d) π/2

Q4. For a unit vector p, if $(\mathbf{x} - \mathbf{p}) \cdot (\mathbf{x} + \mathbf{p}) = 24$, find the value of $|\mathbf{x}|$:

(a) 4 (b) 5

(c) √24 (d) 25

Q5. Vector Projection Calculations:

(i) Find the projection of $\mathbf{a} = 3\mathbf{i} - \mathbf{j} + 4\mathbf{k}$ on $\mathbf{b} = 2\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}$

(ii) Find the projection of $\mathbf{a} = 4\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$ on $\mathbf{b} = 3\mathbf{i} - 2\mathbf{j} + \mathbf{k}$

(iii) Find the projection of $\mathbf{a} = 2\mathbf{i} + 5\mathbf{j} + \mathbf{k}$ on $\mathbf{b} = \mathbf{i} - 2\mathbf{j} + 2\mathbf{k}$

Q6. Cross Product & Perpendicular Vectors:

(i) Find a unit vector perpendicular to $(\mathbf{a} + \mathbf{b})$ and $(\mathbf{a} - \mathbf{b})$, where $\mathbf{a} = 2\mathbf{i} + \mathbf{j} + \mathbf{k}$ and $\mathbf{b} = \mathbf{i} + 3\mathbf{j} - \mathbf{k}$

(ii) Find a unit vector perpendicular to $(\mathbf{a} + \mathbf{b})$ and $(\mathbf{a} - \mathbf{b})$, where $\mathbf{a} = 4\mathbf{i} - \mathbf{j} + 3\mathbf{k}$ and $\mathbf{b} = 2\mathbf{i} + 2\mathbf{j} - \mathbf{k}$

Q7. Compute the area of a triangle whose adjacent sides are determined by vectors $\mathbf{a} = \mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$ and $\mathbf{b} = 2\mathbf{i} + 3\mathbf{j} - \mathbf{k}$.

UNIT 2: THREE DIMENSIONAL GEOMETRY

Q8. Given Cartesian line equation: $(x - 1) / 4 = (3y - 6) / (-6) = z + 2$. If its vector representation is $\mathbf{r} = (\mathbf{i} + 2\mathbf{j} - 2\mathbf{k}) + \lambda(4\mathbf{i} - 2\mathbf{j} + m\mathbf{k})$, determine m:

(a) 1 (b) 0

(c) −1 (d) 2

Q9. Determine the direction cosines of lines passing through the given points:

(i) Point P(1, −1, 2) and Point Q(3, 4, −2)

(ii) Point P(5, 2, −4) and Point Q(−1, 6, 2)

(iii) Point P(−3, 2, 1) and Point Q(1, −4, 5)

Q10. Calculate the shortest distance between the following pairs of skew lines:

(i) $\mathbf{r} = (4\mathbf{i} - \mathbf{j}) + \lambda(\mathbf{i} + 2\mathbf{j} - 3\mathbf{k})$ and $\mathbf{r} = (\mathbf{i} - \mathbf{j} + 2\mathbf{k}) + \mu(2\mathbf{i} + 4\mathbf{j} - 5\mathbf{k})$

(ii) $(x - 1)/2 = (y - 2)/3 = (z - 3)/4$ and $(x - 2)/3 = (y - 4)/4 = (z - 5)/5$

(iii) $\mathbf{r} = (2\mathbf{i} - 3\mathbf{j} + 4\mathbf{k}) + \lambda(3\mathbf{i} + 4\mathbf{j} - 5\mathbf{k})$ and $\mathbf{r} = (\mathbf{i} + \mathbf{j} - \mathbf{k}) + \mu(2\mathbf{i} - 3\mathbf{j} + \mathbf{k})$

UNIT 3: LINEAR PROGRAMMING (LPP)

Q11. The corner points of the feasible region for an LPP are A(0, 4), B(2, 5), C(5, 2) and D(6, 0). The maximum value of objective function Z = 4x + 3y occurs at:

(a) A (b) B

(c) C (d) D

Q12. The corner points of the bounded feasible region for an LPP are P(0, 6), Q(2, 4), R(4, 1) and S(8, 0). The minimum value of Z = 3x + 4y is obtained at:

(a) P (b) Q

(c) R (d) S

Q13. Graphical LPP Optimization:

(i) Maximise $Z = 5x + 3y$ subject to constraints: $3x + 5y \leq 15$, $5x + 2y \leq 10$, $x \geq 0$, $y \geq 0$.

(ii) Minimise $Z = 3x + 9y$ subject to constraints: $x + 3y \geq 6$, $x + y \geq 4$, $x \geq 0$, $y \geq 0$.

(iii) Maximise and Minimise $Z = x + 2y$ subject to: $x + 2y \geq 100$, $2x - y \leq 0$, $2x + y \leq 200$, $x \geq 0$, $y \geq 0$.

UNIT 4: PROBABILITY

Q14. If P(A) = 0.6, P(B) = 0.3 and P(A ∩ B) = 0.18, evaluate conditional probability P(A/B):

(a) 0.3 (b) 0.6

(c) 0.5 (d) 0.2

Q15. Independent Event Problems:

(i) If E and F are independent events with P(E) = 0.35 and P(F) = 0.45, find P(E/F) and P(E ∩ F).

(ii) Two independent events X and Y have P(X) = 0.4 and P(X ∪ Y) = 0.7. Find P(Y).

(iii) A and B solve a problem independently with probabilities 1/2 and 1/3 respectively. Find the probability that the problem is solved.

Q16. CASE STUDY PROBLEM 1: Industrial Component Inspection

A manufacturing company operates three assembly units U_1, U_2, and U_3 producing identical machine parts in proportions 5 : 3 : 2. Defect rates for components from U_1, U_2, and U_3 are known to be 2%, 4%, and 5% respectively.

(i) Calculate the overall probability that a component selected at random from the output is defective.

(ii) Given that a randomly inspected component is defective, find the probability that it came from unit U_1.

— OR —

(ii) Given that a randomly inspected component is defective, find the probability that it came from unit U_3.

Q17. CASE STUDY PROBLEM 2: Diagnostic Test Evaluation

In a given population, 1% of individuals suffer from a specific viral infection. A diagnostic test correctly detects the infection 99% of the time for infected persons, but yields a false positive result for 0.5% of healthy individuals tested.

(i) Find the total probability that a randomly chosen person tests positive.

(ii) If a person tests positive, find the probability that the individual actually has the infection.

— OR —

(ii) If a person tests negative, calculate the probability that the individual is actually healthy.

Q18. CASE STUDY PROBLEM 3: Medical Treatment Outcome

Patients with a certain illness are treated with standard medication or a new therapy option. 60% opt for standard medication while 40% choose the new therapy. The success rates are 70% for standard medication and 85% for the new therapy.

(i) Find the probability that a patient chosen at random recovers from the illness.

(ii) If a patient recovered, what is the probability that they received the new therapy?